

AgriSense — An ML-Based Agricultural Decision Support System for Small-Scale Farmers

Project-Based Learning · Machine Learning · B.E. Computer Science and Engineering · Academic Year 2026–2027

PROJECT TEAM

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FACULTY GUIDE

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01 ABSTRACT

Small-scale farmers choose crops and judge expected harvests from experience rather than data. **AgriSense** pairs two supervised models — a Random Forest recommending one of 22 crops from soil and weather readings, and an XGBoost regressor estimating yield per hectare — behind a FastAPI service and a React Native application. Trained on 2,200 and 48,681 public records, it reaches **99.55%** classification accuracy and **R² 0.7403**, answering both questions on a phone in under a second.

02 PROBLEM STATEMENT

- Most Indian holdings are small or marginal; crop choice and yield expectation rest on inherited experience.
- Soil tests return **nutrient values, not decisions** — the gap is interpretation, not measurement.
- Shifting rainfall and repeated cropping erode the reliability of that experience.
- Prior ML work rarely reaches the farmer: it ends at a notebook, not a usable tool.

03 OBJECTIVES

- Preprocess two public datasets; derive yield and remove invalid and extreme records.
- Build and iteratively refine classification and regression models.
- Benchmark three models per task against a baseline, then tune the winner.
- Evaluate with accuracy / precision / recall / F1 and RMSE / MAE / R².
- Deliver the trained models through a REST API and a mobile client.

04 DATASETS

Crop Recommendation [4]

2,200 rows · 7 numeric features (N, P, K, temperature, humidity, pH, rainfall) · **22 balanced classes**, 100 records each · stratified 80/20 → 1,760 / 440

Indian Crop Production [5]

49,999 rows → **48,681 after cleaning** · 75 crops · 6 seasons · target **Yield = Production ÷ Area** · clipped to the 1st–99th percentile (0.0005–33,089 → 0.168–129.2) · 15,000 sampled → 12,000 / 3,000

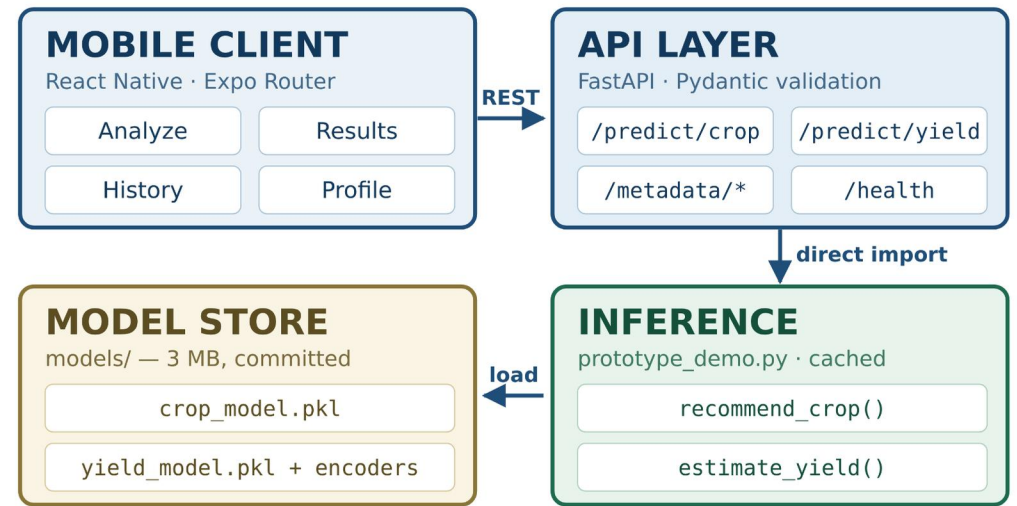
10 CONCLUSION

AgriSense delivers crop recommendation and yield estimation from measurements a small-scale farmer can realistically obtain. A tuned Random Forest classifies 22 crops at **99.55%** test accuracy (5-fold CV 0.9927 ± 0.0039), and a tuned XGBoost regressor estimates yield at **R² 0.7403** (RMSE 5.0247). Both are served through a REST API and a React Native application, with an automated parity test asserting that what the app displays matches the model exactly.

05 METHODOLOGY



06 SYSTEM ARCHITECTURE



Three layers with the ML layer as the single source of truth. The API **imports** the inference module rather than reimplementing it — a parity test asserts endpoint output equals direct model output to 1e-5. The .pkl artefacts are produced offline by the training pipelines (panel 05) and committed, so nothing retrains at request time.

07 IMPLEMENTATION & TECHNOLOGIES

- Training pipelines** — clean, encode, compare three models, tune the winner, serialise with joblib; all metrics written to JSON.
- Inference module** — two functions, models cached after one load; unknown crop or season raises a readable error mapped to HTTP 422.
- REST service** — predict, metadata and health endpoints; dropdowns populated from the saved encoders, so an unknown crop can never be offered. Validated by **22 automated tests** (9 ML + 13 API) and static type checking.



11 FUTURE ENHANCEMENTS

- Train on all 48,681 cleaned rows instead of a 15,000-row sample.
- Replace the coarse integer weather columns with a meteorological API feed.
- Return class probabilities and prediction intervals as a confidence measure.
- Deploy with authentication and a restricted cross-origin policy.
- Add offline support and regional-language interface options.

08 RESULTS & PERFORMANCE

99.55%

Test accuracy
Crop recommendation
CV 0.9927 ± 0.0039

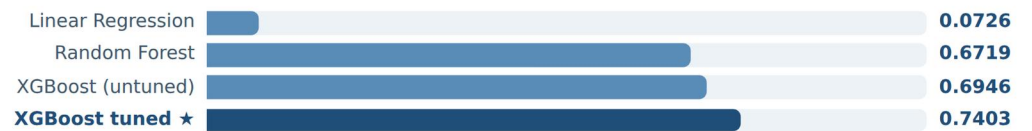
0.7403

R² — yield estimation
RMSE 5.0247 · MAE 1.6527
CV 0.7428 ± 0.0254

Classification — test accuracy (440 held-out records)



Regression — R² (3,000 held-out records)



Final model

Random Forest

Hyperparameters (grid search)

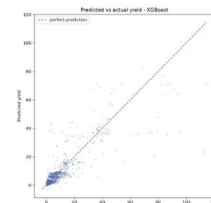
100 trees · depth 10 · min_split 5

XGBoost

200 rounds · depth 4 · lr 0.1

Confusion matrix

438 of 440 held-out records correct



Predicted vs actual yield — tight where most data lies, fanning above, where high-yield records are sparse

09 KEY FINDINGS

- The same model class was near-sufficient for one task and useless for the other — logistic regression **97.27%**, but linear regression only **R² 0.0726**. Crop choice turns on continuous water variables (rainfall 0.2231 + humidity 0.2169 = 44%), which a linear boundary can largely capture.
- Yield is driven by categorical inputs — crop **0.4949** + season **0.3404** = **84%** of importance — which a linear model over label-encoded integers cannot represent.
- Tuning could not improve the classifier (2 of 440 errors — the dataset ceiling) but lifted the regressor 0.6946 → **0.7403**, choosing depth 4 over 6 and so correcting overfitting.
- Running the real application exposed **5 integration defects** invisible to the 22 unit tests and to type checking.
- Limitations:** the balanced dataset flatters classification accuracy; coarse integer weather columns and sparse high-yield records cap the regressor; the build is not deployed.

12 REFERENCES

- [1] L. Breiman, “Random forests,” *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.
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- [3] F. Pedregosa et al., “Scikit-learn: Machine learning in Python,” *JMLR*, vol. 12, pp. 2825–2830, 2011.
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- [5] A. Kandoi, “Crop prediction dataset based on Indian agriculture,” GitHub repository.
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